they are likely to succeed: Zeilinger is an experimental master of photon manipulation. Zeilinger's group has already demonstrated that it can send and receive single photons over distances much longer than the 10 kilometers from the lab to the mirror in the vineyard and back. In 2012, they successfully sent entangled photons over 143 kilometers, between La Palma and Tenerife in the Canary Islands. And Zeilinger has also spent decades conducting improved versions of Aspect's Bell experiments, verifying the existence of quantum nonlocality with enormous experimental precision.

Yet despite his intimate knowledge of the most bizarre aspects of the quantum world, Zeilinger has no qualms about the Copenhagen interpretation. "The quantum state, as Heisenberg says, is a mathematical representation of our knowledge," said Zeilinger. "It tells us the set of possible future measurement results, together with their probabilities." Measurement, for Zeilinger, plays a central role in quantum physics. "There is no measurement problem," he claims. "Measurement results live in the classical world, and the quantum state is what we call a quantum world, which is only a mathematical representation, according to Heisenberg.… What you can talk about with your classical language, these are the objectively existing objects of the universe, these are the classical objects. And that's it. That's what can be talked about. The rest is mathematics." In other words, there are two worlds: a world of actually existing everyday objects obeying classical prequantum physics, and a quantum "world" that isn't real in the same way, just as Heisenberg said. But Zeilinger doesn't think that there is a true boundary between them, some line beyond which quantum physics does not apply. "There is no fundamental boundary," he said. "There is a transition from classical to quantum, but that is not a boundary." It's not surprising that Zeilinger would say this: almost no physicists still believe that there is any such fundamental boundary, and some of the most convincing work against such an idea came from Zeilinger himself. Back in 1999, Zeilinger and his collaborators managed to coax a buckyball—a collection of sixty carbon atoms in the shape of a soccer ball—to interfere with itself, like a photon in a double-slit experiment. Finding quantum effects in an object so much larger than an individual subatomic particle (though still about a billion times smaller than the objects in our everyday lives) might have been shocking to some of the